

# Electric Vehicle Data Analysis Project

## Project Overview

In this project, you will analyze a dataset related to electric vehicles (EVs). The dataset contains various features such as electric range, energy consumption, price, and other relevant attributes. Your goal is to conduct a thorough analysis to uncover meaningful insights, tell a compelling story, conduct hypothesis testing and provide actionable recommendations based on the data.

**Dataset:** [FEV-data-Excel.xlsx](#)

## Dataset Overview

**Car full name:** The full name or designation of the vehicle, often combining make, model, and variant.

**Make:** The brand or manufacturer of the car.

**Model:** The specific model or version of the car.

**Minimal price (gross) [PLN]:** The minimum retail price of the car, in Polish złoty (PLN).

**Engine power [KM]:** The car's engine power, measured in horsepower (KM in Polish).

**Maximum torque [Nm]:** The peak torque the engine can produce, measured in Newton-meters (Nm).

**Type of brakes:** The braking system used, such as disc or drum brakes.

**Drive type:** The drivetrain configuration, like FWD (front-wheel drive), RWD (rear-wheel drive), or AWD (all-wheel drive).

**Battery capacity [kWh]:** Total energy capacity of the car's battery, measured in kilowatt-hours (kWh).

**Range (WLTP) [km]:** Estimated driving range on a full charge under WLTP standards, in kilometers.

**Wheelbase [cm]:** The distance between the front and rear axles, in centimeters.

**Length [cm]:** The overall length of the car, in centimeters.

**Width [cm]:** The car's width, in centimeters.

**Height [cm]:** The car's height, in centimeters.

**Minimal empty weight [kg]:** The car's minimum weight when empty, measured in kilograms.

**Permissible gross weight [kg]:** Maximum legally allowed weight, including passengers and cargo, in kilograms.

**Maximum load capacity [kg]:** The maximum weight the car can carry, in kilograms.

**Number of seats:** The number of passenger seats in the car.

**Number of doors:** The number of doors on the car.

**Tire size [in]:** The tire size, measured in inches.

**Maximum speed [kph]:** The top speed of the car, in kilometers per hour.

**Boot capacity (VDA) [l]:** Trunk or cargo space capacity, measured in liters according to VDA standards.

**Acceleration 0-100 kph [s]:** Time taken to accelerate from 0 to 100 kilometers per hour, in seconds.

**Maximum DC charging power [kW]:** The highest charging power supported when using a DC fast charger, in kilowatts (kW).

**Mean - Energy consumption [kWh/100 km]:** Average energy consumption per 100 kilometers, in kilowatt-hours (kWh).

## **Instructions**

- Use Python and libraries such as Pandas, NumPy, SciPy, Matplotlib, and any other tools you find necessary.
- Focus on clear, structured code and explanations to guide readers through your thought process.
- Every task requires both a coding solution and a written analysis section explaining your findings.